# Data Quality Report

AeroGrid | Source: live Neon warehouse | UTC: 2026-11-13T14:20:39.523016+00:00

## Audit summary

DQ run: 7; critical FAIL checks: 0; warning checks: 3. Refer to each check below.

## Check catalog results

| Check ID | Severity | Status | Observed / Expected |
|---|---|---|---|
| DQ-A-001 | CRITICAL | PASS | 0 / 0 |
| DQ-A-002 | CRITICAL | PASS | 0 / 0 |
| DQ-A-007 | CRITICAL | PASS | 0 / 0 |
| DQ-A-008 | WARNING | PASS | 0 / 0 |
| DQ-B-001 | CRITICAL | PASS | 0 / 0 |
| DQ-B-003 | CRITICAL | PASS | 0 / 0 |
| DQ-B-004 | CRITICAL | PASS | 0 / 0 |
| DQ-C-001 | CRITICAL | PASS | 0 / 0 |
| DQ-C-003 | WARNING | WARN | 27479 / 0 |
| DQ-C-005 | CRITICAL | PASS | 0 / 0 |
| DQ-C-006 | CRITICAL | PASS | 0 / 0 |
| DQ-C-007 | WARNING | PASS | 0 / 0 |
| DQ-C-008 | WARNING | WARN | 27479 / 0 |
| DQ-I-001 | CRITICAL | PASS | 0 / 0 |
| DQ-I-002 | CRITICAL | PASS | 0 / 0 |
| DQ-I-003 | CRITICAL | PASS | 0 / 0 |
| DQ-I-004 | CRITICAL | PASS | 0 / 0 |
| DQ-I-005 | CRITICAL | PASS | 0 / 0 |
| DQ-I-006 | CRITICAL | PASS | 0 / 0 |
| DQ-I-008 | CRITICAL | PASS | 0 / 0 |
| DQ-I-009 | CRITICAL | PASS | 0 / 0 |
| DQ-I-010 | CRITICAL | PASS | 0 / 0 |
| DQ-R-001 | CRITICAL | PASS | 0 / 0 |
| DQ-R-002 | CRITICAL | PASS | 0 / 0 |
| DQ-R-004 | CRITICAL | PASS | 0 / 0 |
| DQ-R-005 | CRITICAL | PASS | 0 / 0 |
| DQ-R-006 | CRITICAL | PASS | 0 / 0 |
| DQ-T-003 | WARNING | WARN | 1 / 0 |
| DQ-U-001 | CRITICAL | PASS | 0 / 0 |
| DQ-U-002 | CRITICAL | PASS | 0 / 0 |
| DQ-U-003 | CRITICAL | PASS | 0 / 0 |

| Check ID | Severity | Status | Observed / Expected |
|---|---|---|---|
| DQ-U-004 | CRITICAL | PASS | 0 / 0 |

## Open DQ warnings explained

| Check | Observed | Meaning / source limitation |
|---|---|---|
| DQ-C-003 | 27479 | Unknown weather-key assignments; no matched real METAR weather observation is linked to these tracking facts. |
| DQ-C-008 | 27479 | Source on_ground is NULL; WP-07 airborne eligibility is inferred and does not replace the observed field. |
| DQ-T-003 | 1 | One warning indicator for missing real restriction snapshots; this is NOT one missing flight or one restriction event. |

## Real warehouse row counts

| Schema.table | Rows |
|---|---|
| project_staging.stg_adsb_position | 27479 |
| project_staging.stg_reject | 418 |
| project_staging.stg_airspace_zone | 0 |
| project_staging.stg_restriction_snapshot | 0 |
| project_dw.dim_time | 169 |
| project_dw.dim_site | 5 |
| project_dw.dim_vehicle | 157 |
| project_dw.dim_spatial_grid | 3659 |
| project_dw.dim_weather | 1 |
| project_dw.dim_airspace_zone | 1 |
| project_dw.fact_vehicle_tracking | 27479 |
| project_dw.fact_flight_journey | 178 |
| project_dw.fact_close_encounter | 0 |
| project_dw.fact_airspace_occupancy | 1754 |
| project_dw.fact_restriction_activation | 0 |

## Unknown dimension members

Dimension totals include the mandatory -1 unknown member. For example, a one-row dim_weather does NOT imply one measured weather observation, and a one-row dim_airspace_zone does NOT imply one real FAA restriction.

## Capacity

Actual PostgreSQL database size: 52,781,056 bytes. Object-storage usage is not available from the PostgreSQL connection; do not equate manifest bytes with current bucket usage.

## Encounter and restriction limitations

The WP07-v1 encounter algorithm requires identical timestamps. Near-time diagnostics were separate analytical SQL, not the production algorithm. No historical 2022 TFRs are inferred from 2026 FAA services. Zero derived rows may be legitimate source coverage, not evidence of a missing SQL table.

## Decision rule

Any stored CRITICAL FAIL blocks trusted publication even if this PDF can be generated. A green unit-test run alone does not establish data quality.

## Evidence and reproducibility

Latest ETL run ID: 7, mode=validate, started=2026-11-13 13:19:02.457585+00:00, status=SUCCESS, version=wp08-v1. DQ check IDs map to Phase2/etl/dq.py and Phase2/docs/06_data_quality_and_reports.md. Verify the source manifests and rejected-row counts directly in Neon.